

Code 1 - Virtual

```
mahij@Mahija: ~  
mahij@Mahija:~$ nano virtual.c  
mahij@Mahija:~$ gcc virtual.c -o virtual  
mahij@Mahija:~$ ./virtual  
Value of x = 10  
Virtual address of x = 0x7ffedf7813a4  
mahij@Mahija:~$ |
```

Code 2 - Address Space

```
mahij@Mahija:~$ nano address space.c
mahij@Mahija:~$ nano addressspace.c
mahij@Mahija:~$ gcc addressspace.c -o addressspace
mahij@Mahija:~$ ./addressspace
Code address : 0x5d296b1e91a9
Data address : 0x5d296b1ec010
BSS address  : 0x5d296b1ec018
Heap address : 0x5d298dfab010
Stack address: 0x7ffcb5fec70c
mahij@Mahija:~$ |
```

Code 3 - Demand Paging

```
mahij@Mahija:~$ nano demandpage.c
mahij@Mahija:~$ gcc demandpage.c -o demandpage
mahij@Mahija:~$ ./demandpage
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
mahij@Mahija:~$ |
```

Code 4 - Address Translation

```
mahij@Mahija:~$ nano addtrans.c
mahij@Mahija:~$ gcc addtrans.c -o addtrans
mahij@Mahija:~$ ./addtrans
Virtual Address = 250
Page Number     = 2
Offset          = 50
Frame Number    = 1
Physical Address = 150
mahij@Mahija:~$ |
```

Code 5 - Memory

```
mahij@Mahija:~$ nano memory.c
mahij@Mahija:~$ gcc memory.c -o memory
mahij@Mahija:~$ ./memory
10 20 30 40 50
mahij@Mahija:~$ |
```

Code 6 - Access Memory

```
mahij@Mahija:~$ nano accessmemory.c
mahij@Mahija:~$ gcc accessmemory.c -o accessmemory
mahij@Mahija:~$ ./accessmemory
Before accessing memory
Segmentation fault      ./accessmemory
mahij@Mahija:~$ |
```